Thomas Li

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PROFESSIONAL SUMMARY

Highly motivated UC Berkeley Math/CS student specializing in software engineering and statistical ML, with hands-on industry and research experience spanning applied mathematics, machine learning, and international academic institutions.

EDUCATION

UC Berkeley | GPA: 4.0/4.0 | B.A. Applied Mathematics

Entered with full completion of general education requirements and all lower-division coursework, freeing coursework capacity for advanced Math, Econ, and CS electives

Santa Rosa Junior College | GPA: 4.0/4.0 | Phi Theta Kappa Honor Society | 2024-2026

- Completed IGETC and core major requirements for both Computer Science and Mathematics programs, while a senior in high school

Maria Carrillo High School | GPA 4.94/5.0, Rank 3/404 | National Merit Finalist | SAT: 1560/1600

PROFESSIONAL EXPERIENCE

Software Engineering & ML Intern | Keysight Technologies | Santa Rosa, CA | June 2026 - Present

Agentic AI & Test Automation

- Developed and validated agentic AI workflows for test measurement engineering, boosting hardware engineering productivity through plugins and MCP integrations that encode engineering team conventions as reusable skills.
- Helped build test measurement libraries in C#/.NET for Keysight instrument testing, gaining hands-on experience with unit testing, NuGet packaging, and Jenkins-based CI/CD deployment

Anomaly Detection & Machine Learning

- Co-developing a statistical ML anomaly detection model to flag defective units of the Keysight M9484C VXG (54 GHz Vector Signal Generator) during production calibration, reducing test cycle time by catching failures early on instruments valued at $1M+

TECHNICAL SKILLS

- **Languages:** C++, C#, Python, Java, Assembly (x86, ARM)
- **Platforms & OS:** Windows, Linux
- **Software & Tools:** Visual Studio, PyCharm, Jupyter Notebooks, GDB, Keysight Advanced Design System (ADS), Git
- **Engineering Practices:** Unit Testing, NuGet Packaging, Jenkins CI/CD, .NET Architecture
- **Methodologies:** OOP, Template Programming, Statistical Machine Learning

RESEARCH EXPERIENCE & PROJECTS

Physics-Informed Machine Learning Research Intern | UCSB Summer Research Academies

- Predicted start-up sentiment trends using Dynamical Systems Theory, Piecewise Deterministic Markov processes, and physics-informed machine learning; co-authored and presented research paper at UCSB conference under guidance from PhD student Charles Johnson, Mechanical Engineering.

Financial Modeling & Deep Learning Project | Imperial College London

- Combined sentiment analysis (FinBERT/LGBM), statistical time series forecasting (ARIMA/LSTM), and financial report data through a deep neural network ensemble using Python and Colab Notebooks under mentorship of Professor Adam Almakroudi.

TinyML Mentorship | KAUST

- Developed TinyML and computer vision projects in motion, object, and anomaly detection using Arduino NICLA Vision ABX00051, C++, Edge Impulse, and Python. Performed real-time inference on memory-limited hardware. Hands-on experience with microcontrollers and embedded systems under mentorship of Prof. Amanat Kafizov

Mathematics Research | SRJC Mathematics Department

- Applied integral transforms to develop a novel framework for indefinite summation, enabling analytical analysis of difference equations. Self-initiated research with SRJC professors; paper (ResearchGate) under review at Rose-Hulman's Undergraduate Mathematics Journal

Rough Path Theory Research | SRJC

- Developed accessible explanations and applications of Rough Path Theory for time series analysis and machine learning under Prof. Lori Lewis; presented at SRJC Research Showcase.

C++ Algorithm Design & Data Structures Portfolio

- **GitHub** portfolio of advanced OOP, data structures, algorithm optimization, and template design projects in Modern C++, reflecting strong software engineering fundamentals applicable to embedded systems and tooling.

LEADERSHIP, SERVICE, AND HOBBIES

Head Coach & Mentor | MATHCOUNTS (Rincon Valley MS/MCHS)

- Competitively selected to lead 60+ students in grades 5-8; developed and delivered weekly curricula; consistently qualified 10-15 students to State competition annually.
- Tracked individual student progress through practice exams and detailed performance sheets, informing coaching decisions, communication, and organization.

Warehouse Volunteer | Redwood Empire Food Bank

- Organized warehouse food stock for distribution, working in distribution at the Food Connections Market, serving community members in food insecurity.

Competition Piano | Music Teacher's Association of California (MTAC)

- Competed in Certificate of Merit for 5+ years; Placed 1st/2nd locally (2019-2020), continuing recreationally

Advanced Calisthenics

- Passionate about freestyle calisthenics and weightlifting, building physical and mental resilience.